

Frequency Tables & Dot Plots

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS You can show one data set three ways. **Tally marks** record data as you collect it; each mark is one, and the fifth mark crosses the group of four. A **frequency table** lists each choice with a number that tells how many times it happened. A **dot plot** stacks one dot for each item above a number line. Read the **shape**: the tallest stack is the most common, and a gap means no data there.

Round 1 — Retrieve It

1. In tally marks, how do you show the number five?

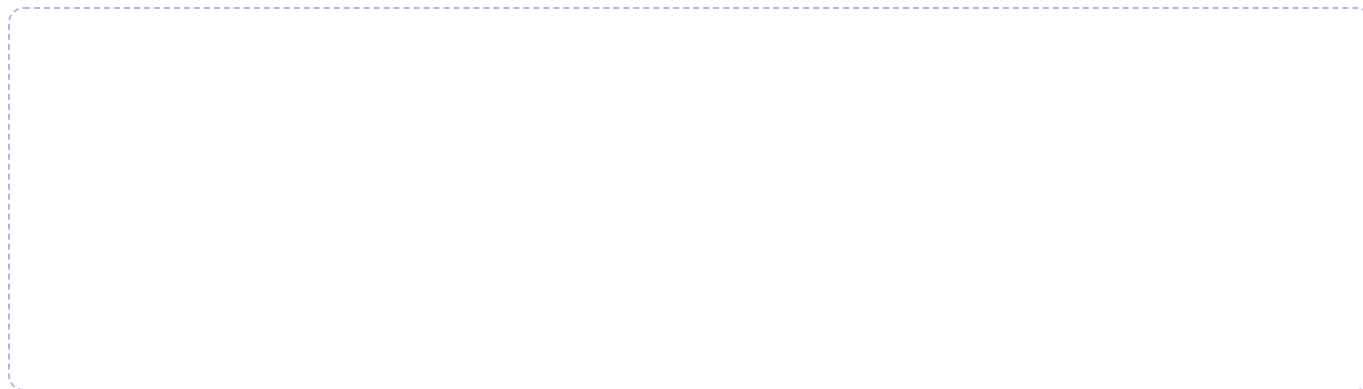
2. What does the number in a frequency table tell you?

3. On a dot plot, what does one dot stand for?

4. On a dot plot, how do you find the most common value?

Round 2 — Sketch It

These are pets per family: 2, 1, 2, 3, 2, 1. Draw a dot plot for the numbers 1, 2, and 3, with one dot for each family.



Frequency Tables & Dot Plots — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A class rolled a number cube. The tally for “3” looks like this: four marks with one mark crossing them, then one more mark. How many times did they roll a 3?

2. A frequency table shows Red 6, Blue 4, Green 2. Make a total. How many votes were there in all?

3. A dot plot of ages has: age 7 with 2 dots, age 8 with 5 dots, age 9 with 1 dot. Which age is most common, and how many children in all?

Round 4 — Reason from Evidence

A group counted books read this week: 3, 3, 4, 3, 5, 3. They showed it as tally marks, a frequency table, and a dot plot. Explain why all three show the same thing, and tell what the shape of the dot plot tells you about the number 3.

Frequency Tables & Dot Plots — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how tally marks, a frequency table, and a dot plot are alike and different. Use the words count, number, and dots.

C**Connect**

Connect this to data you could collect in class, such as favorite lunch or shoe size. Tell which of the three ways you would use to record it as you go.

E**Extend**

Take this data set: 5, 6, 6, 6, 7. Turn it into a frequency table, then describe the shape of its dot plot in words.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Frequency Tables & Dot Plots

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Draw four marks and cross them with a fifth mark (a group of five).
2. How many times that choice happened (its count or frequency).
3. One item or one data point (such as one person or one thing).
4. Find the number with the tallest stack of dots.

Sketch It — look for

A number line with 1, 2, 3. Above 1: 2 dots; above 2: 3 dots; above 3: 1 dot. Six dots in all, matching the data.

Apply It

1. 6 times (a group of five plus one more).
2. 12 votes in all ($6 + 4 + 2$).
3. Age 8 is most common; 8 children in all ($2 + 5 + 1$).

Reason from Evidence — look for

All three record the same six values, just in different forms, so the counts match. The dot plot has its tallest stack at 3 (four dots), which shows 3 is the most common number of books.

ACE — Extend look-for

Frequency table: $5 \rightarrow 1$, $6 \rightarrow 3$, $7 \rightarrow 1$. The dot plot has one dot at 5, a tall stack of three at 6, and one dot at 7, so it peaks in the middle at 6.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.